Trend & Spectral Analysis of Croatian Real GDP

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1 Executive Summary

Croatia's economy has grown considerably since the mid-1990s, but not smoothly. This report examines quarterly real GDP from 1995 Q1 to 2026 Q2 (126 quarters) with four methods (polynomial and broken trends, the Hodrick-Prescott filter, spectral analysis and ARFIMA models) to answer three practical questions:

What does the long-run growth path look like, and how did major crises change it? Is there predictable structure in quarterly GDP growth, or does it behave like random noise? Do economic shocks fade out, or do they leave a permanent mark?

- Croatia grew strongly until 2008. The **Global Financial Crisis changed the trajectory for a decade**: the pre-crisis peak of 2008 Q2 was only regained in 2018 Q4, and trend growth fell from 0.80% to only 0.01% per quarter.
- The **COVID-19 shock** produced by far the largest fall in the sample (-14.1% in 2020 Q2). This time the recovery was fast: GDP was back at its pre-pandemic level in 2021 Q1.
- **Quarter-to-quarter growth is not predictable.** Growth rates show no significant autocorrelation: last quarter's growth says nothing about next quarter's.
- **Shocks are permanent, and the 2008 crisis also lowered average growth.** Log GDP behaves like a random walk with drift, so every shock shifts the level of output for good. The drift itself was not constant: average quarterly growth fell from 0.88% before 2009 to 0.13% in 2009 to 2019, which is why the crisis cost Croatia roughly a decade of output growth.

About this Report

Data source	FRED, series CLVMNACSCAB1GQHR
Period	1995 Q1 to 2026 Q2 (quarterly, 126 observations)
Methods	Polynomial and broken trends, HP filter, spectral analysis, ARFIMA
Tools	R (`mFilter`, `fracdiff`, `kableExtra`)
Author	Filip Popovic

2 Data

2.1 The Raw Series

The data come from Federal Reserve Economic Data (FRED), series CLVMNACSCAB1GQHR: Croatian real GDP in chained national currency units, seasonally adjusted (retrieved 6 November 2026). All analyses use the natural logarithm of GDP, so that differences can be read as growth rates.

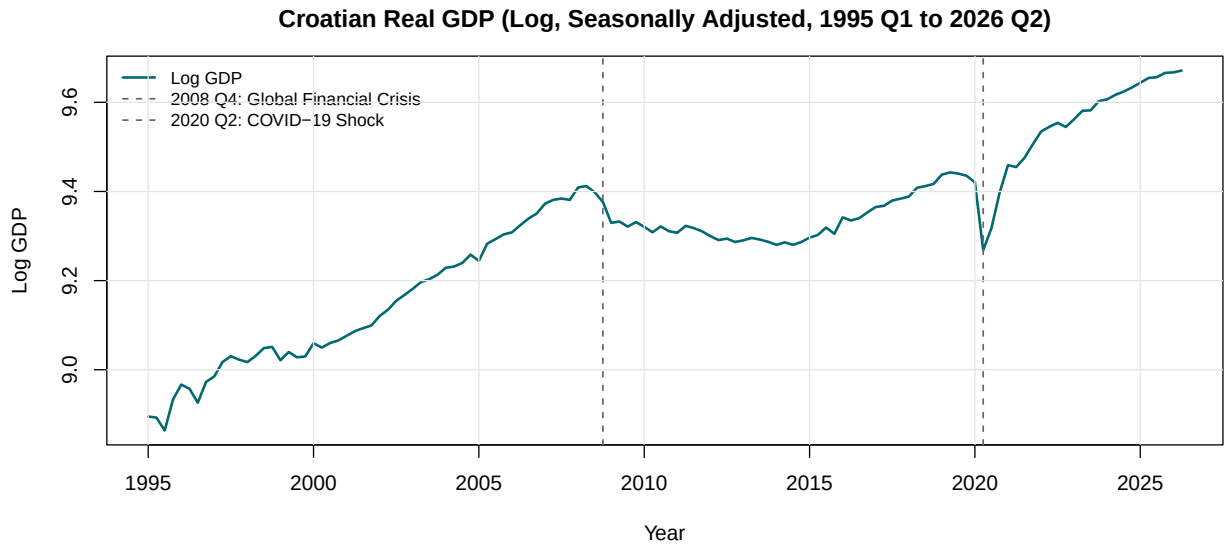


Figure 1: Croatian real GDP (log scale). The dashed vertical lines mark the onset of the Global Financial Crisis (2008 Q4) and the COVID-19 shock (2020 Q2).

The log GDP series shows a clear upward trend with two sharp interruptions. After peaking in 2008 Q2, output contracted during the Global Financial Crisis and stagnated for years. The pre-crisis peak was only regained in 2018 Q4, 10.5 years later. The COVID-19 shock caused a much deeper but short-lived fall: GDP dropped by 14.1% in 2020 Q2 and was back at its 2019 Q4 level by 2021 Q1. A likely driver of this fast rebound is the recovery of tourism, which plays a large role in the Croatian economy. The series is clearly non-stationary and trend-driven, which motivates the detrending methods below.

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3 Trend Analysis

3.1 Polynomial Trends

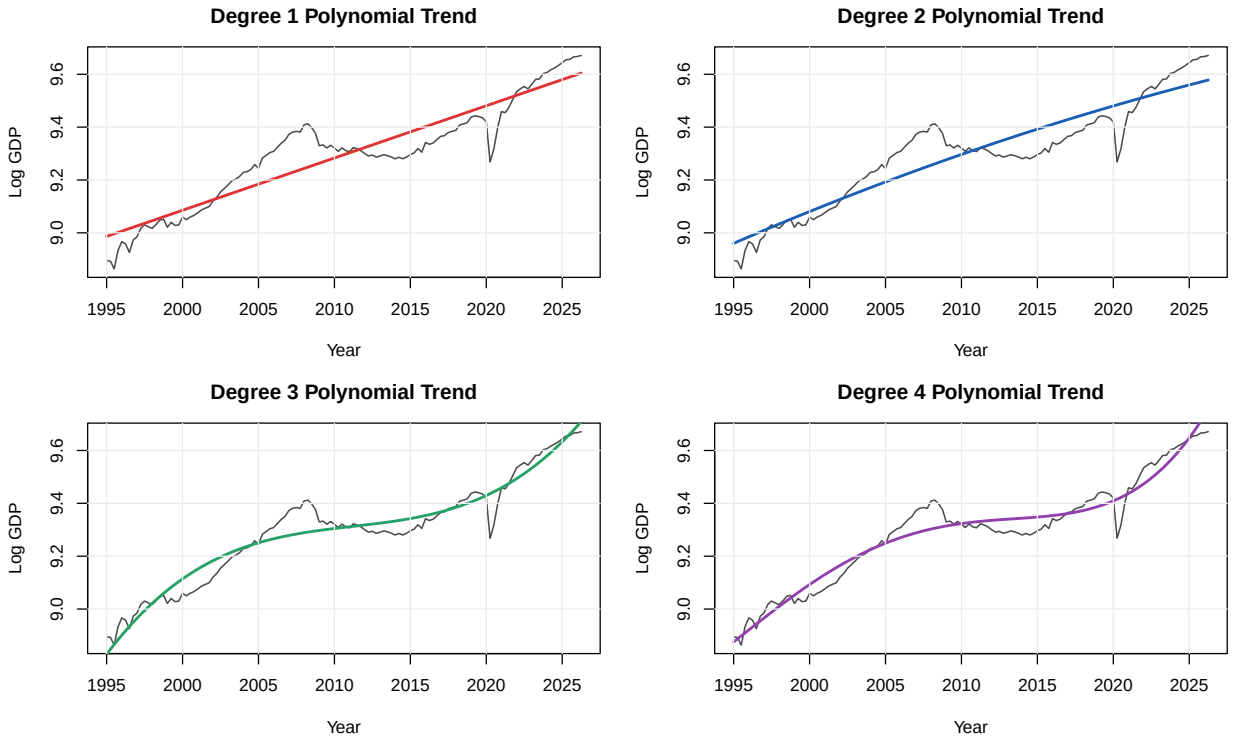


Figure 2: Polynomial trend fits of degrees 1 through 4 overlaid on log GDP. Higher-degree polynomials follow the data more closely but risk overfitting and are sensitive to behaviour at the endpoints of the sample.

Polynomial trends of degrees 1 to 4 are fitted to log GDP by OLS. The linear trend captures the general upward direction but misses the long stagnation after 2008 and the strong growth after 2020. Higher degrees follow the crisis episodes more closely, and the residual sum of squares falls from 0.647 (degree 1) to 0.252 (degree 4). The price for this flexibility is a trend that has no economic meaning and reacts strongly to the endpoints of the sample, a well-known weakness of global polynomial detrending.

3.2 Deviations from Polynomial Trends

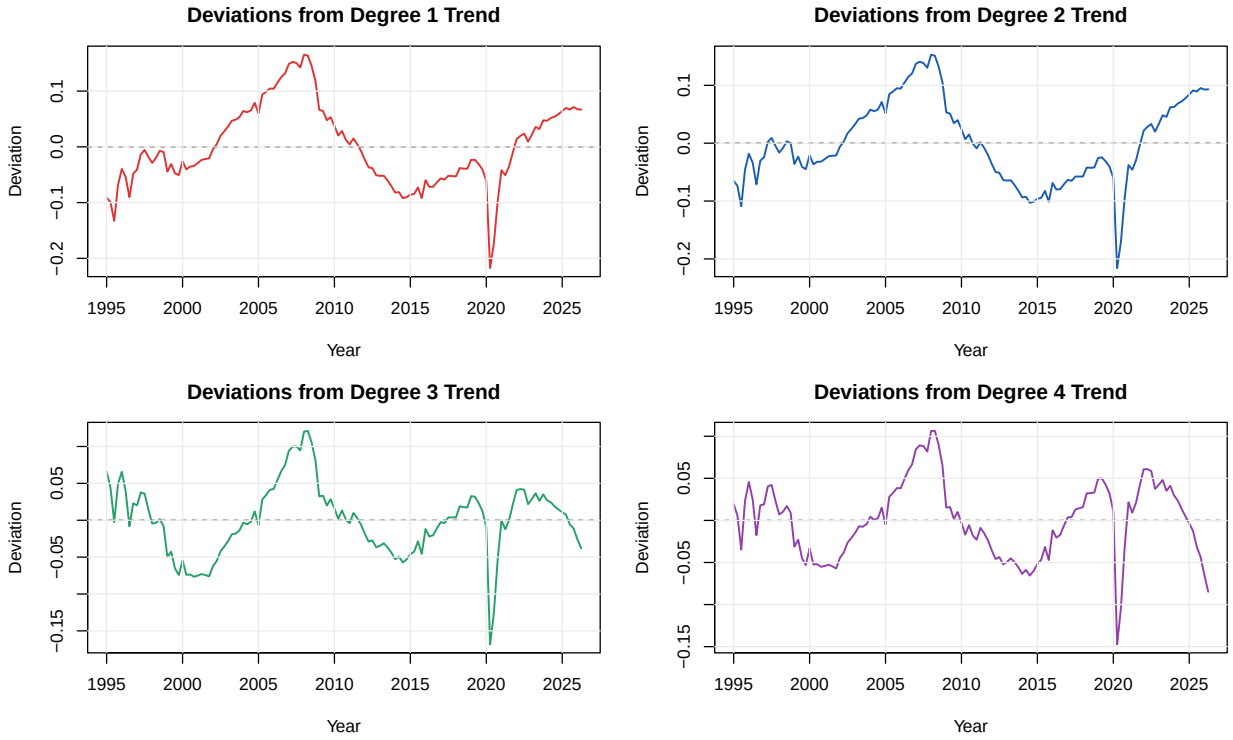


Figure 3: Residuals from polynomial trends of degrees 1 through 4. A well-fitting trend would leave structureless deviations, but all four panels show long, wave-like swings around the two crisis episodes.

At every degree the deviations move in long swings rather than random noise. Even for the degree-4 trend, the first-order autocorrelation of the residuals is 0.86. The GDP path shifted after the two crises in a way that a single smooth polynomial cannot describe, which motivates modelling the breaks explicitly.

3.3 Broken Linear Trends

Following the structure of the course assignment, three broken trend models are compared, with breaks at **2008 Q4** (Global Financial Crisis) and **2020 Q2** (COVID-19):

(a) a break in slope after 2008 Q4,
(b) breaks in slope after 2008 Q4 and 2020 Q2,
(c) breaks in slope after 2008 Q4 and 2020 Q2 and an additional break in the intercept after 2020 Q2.

A break in slope lets the trend growth rate change, while a break in intercept allows the trend to restart at a different level after the break.

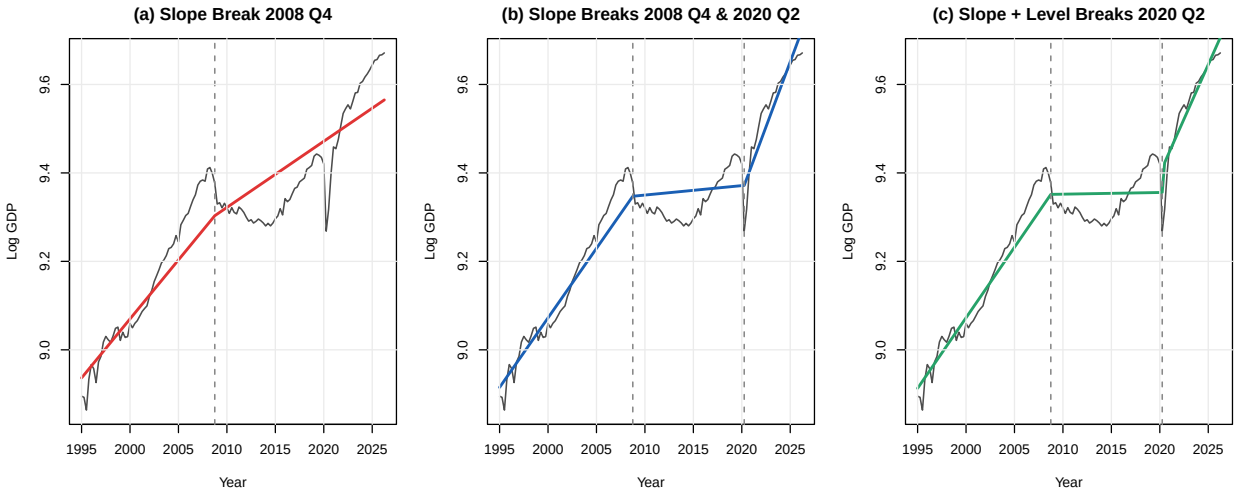


Figure 4: Broken linear trend models. Model (c) additionally allows the trend to restart at a different level after 2020 Q2.

Table 1: Residual sum of squares (RSS) of all trend models.

Model	Parameters	RSS
Polynomial, degree 1	2	0.647
Polynomial, degree 2	3	0.628
Polynomial, degree 3	4	0.287
Polynomial, degree 4	5	0.252
(a) Slope Break 2008 Q4	3	0.561
(b) Slope Breaks 2008 Q4 & 2020 Q2	4	0.239
(c) Slope + Level Breaks 2020 Q2	5	0.227

Model (c) fits best of all trend models (RSS 0.227, compared with 0.252 for the degree-4 polynomial) while keeping a clear economic interpretation. It separates three growth phases: trend growth of 0.80% per quarter before the crisis, only 0.01% per quarter between 2009 and 2020, and 1.23% per quarter after 2020. The level break after 2020 Q2 is estimated at 5.5%. It is positive, so the post-pandemic trend starts above the extrapolated pre-pandemic path, reflecting the strong recovery. The COVID-19 drop itself lasted only one quarter and therefore shows up as a deviation from the trend rather than as a shift of it. The Global Financial Crisis, in contrast, did not just cause a temporary dip: it lowered Croatia's trend growth for more than a decade.

3.4 Deviations from Broken Linear Trends

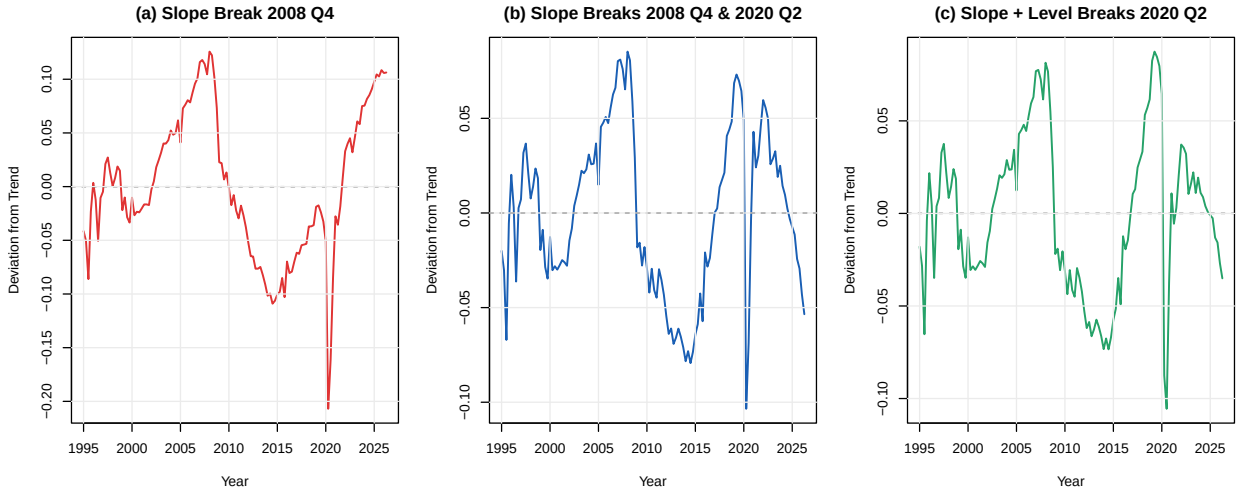


Figure 5: Residuals from the broken trend models. Even the best model leaves long, persistent swings, which shows that the Croatian business cycle is prolonged rather than short-lived.

The deviations from model (c) are the smallest of all trend models, but they are far from random: their first-order autocorrelation is 0.87. Periods above and below trend last for years. This persistence is exactly what one would expect if the level of GDP itself wanders like a random walk, a point the spectral and ARFIMA analysis below will confirm.

3.5 Hodrick-Prescott Filter

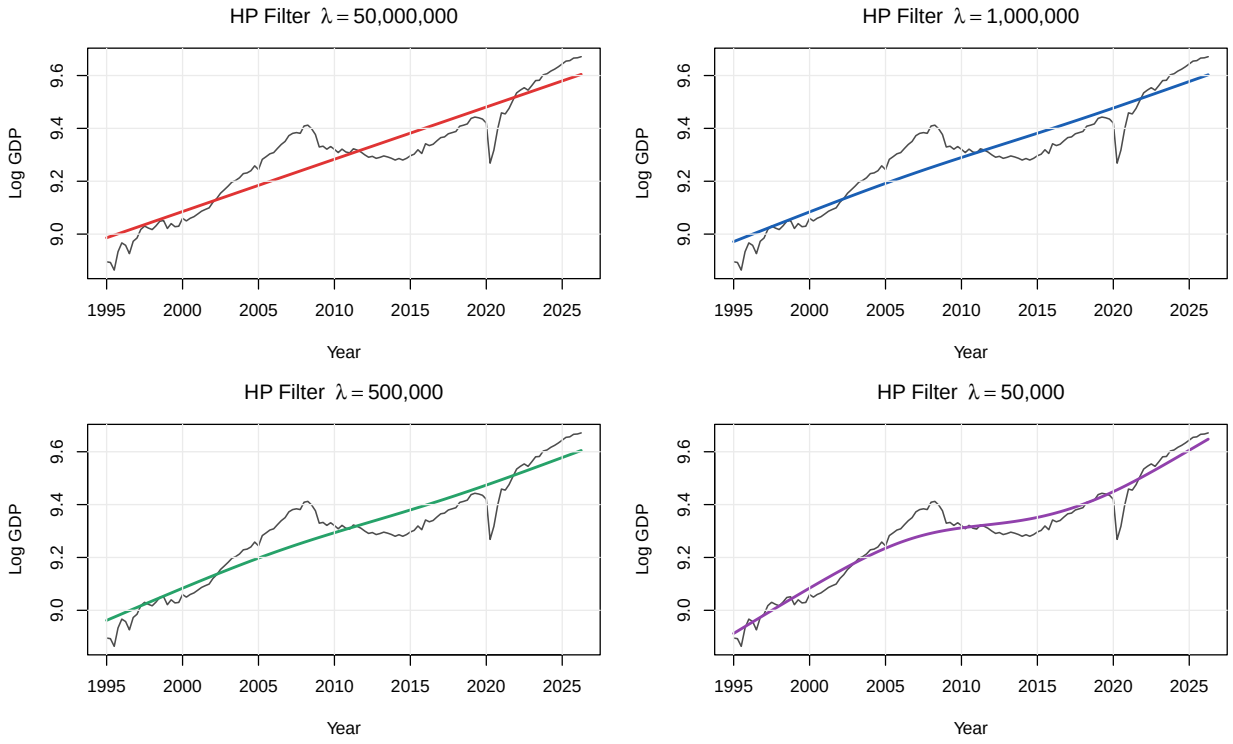


Figure 6: HP filter trend estimates for the four smoothing parameters used in the course. Higher values of lambda produce a stiffer trend; lower values let the trend follow the data more closely.

The Hodrick-Prescott (HP) filter splits GDP into a smooth trend and a cyclical component by minimising a penalised sum of squares. The penalty λ controls the smoothness: at $\lambda = 50{,}000{,}000$ the trend is almost a straight line, while at $\lambda = 50{,}000$ it bends with the crisis episodes and absorbs part of the stagnation after 2008.

3.6 HP Cycle Components

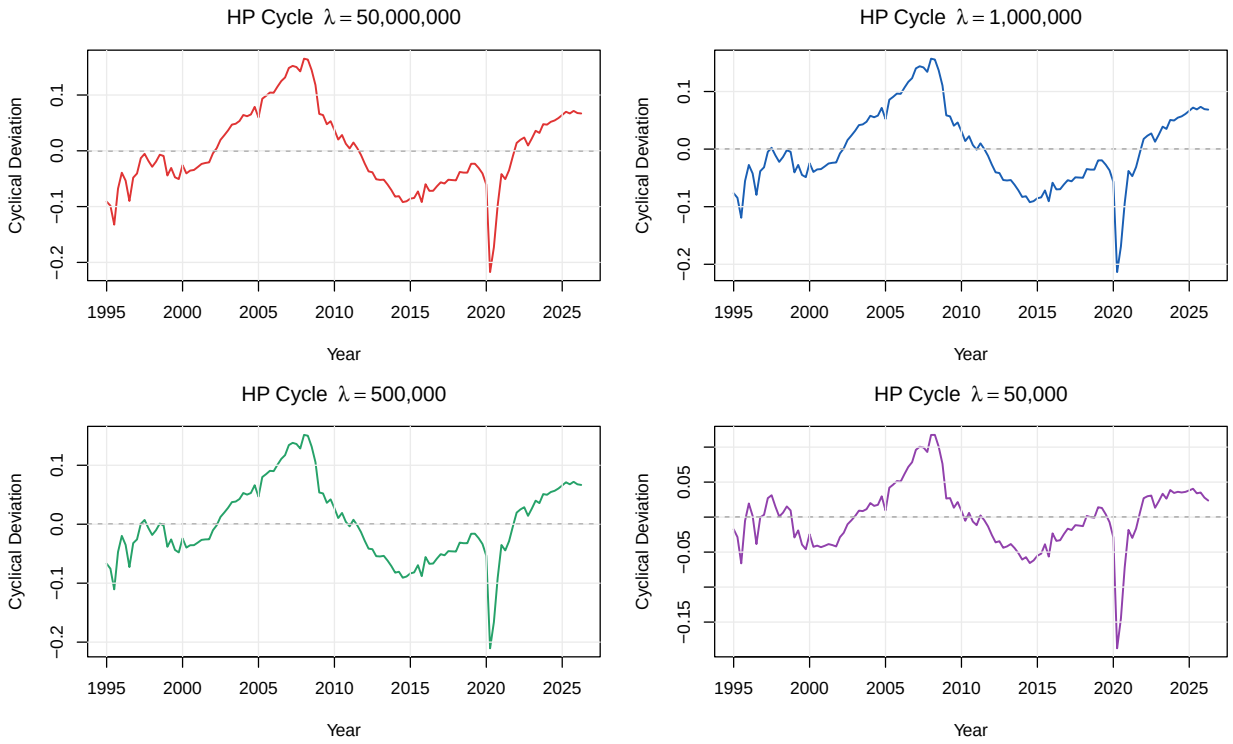


Figure 7: Cyclical components extracted by the HP filter at four smoothing levels. The long below-trend phase after the financial crisis and the COVID-19 dip are visible in all specifications.

The cyclical components tell the same story at every smoothing level: an expansion until 2008, a long period of below-trend output in the 2010s, and the sharp COVID-19 dip followed by a strong rebound. How long the weak phase lasts depends on λ: with $\lambda = 500{,}000$ output stays below trend from 2011 Q3 to 2021 Q4 (42 consecutive quarters), with $\lambda = 50{,}000$ from 2011 Q3 to 2018 Q1 (27 quarters). This sensitivity is a reminder that the HP cycle is a descriptive tool and its exact dating should not be over-interpreted.

3.7 Quarterly Growth Rates

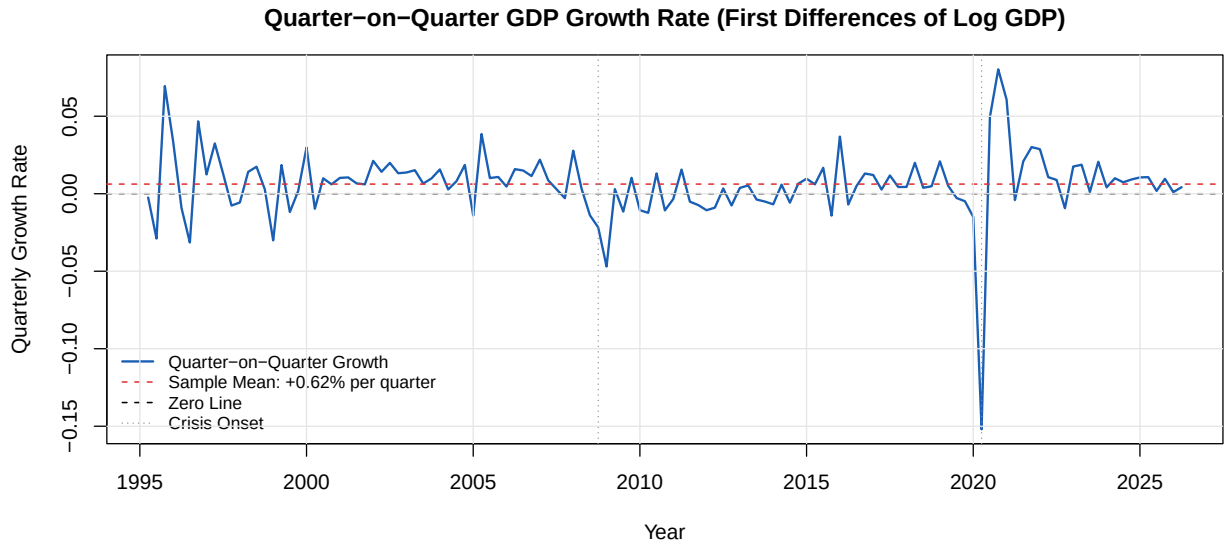


Figure 8: Quarter-on-quarter growth rates of Croatian real GDP (first differences of log GDP). The largest fall is -14.1% in 2020 Q2, the strongest quarter is +8.4% in 2020 Q4.

First differences of log GDP are quarter-on-quarter growth rates. They fluctuate around a positive mean of 0.62% per quarter. For small values a log difference is almost the same as a percentage change; for the large crisis quarters the text converts them to percentage changes. The two crises stand out: -4.6% in 2009 Q1 and -14.1% in 2020 Q2, followed by +5.1% in 2020 Q3 and +8.4% in 2020 Q4. Growth is also more volatile around both crises. The level of growth is not constant either: the mean was 0.88% per quarter before 2009, only 0.13% from 2009 to 2019 and 0.91% from 2020 onwards. Apart from this shift the differenced series looks stationary, which is consistent with log GDP being integrated of order one. The next section checks whether these growth rates contain any predictable structure.

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4 Spectral Analysis

Spectral analysis asks: *at which frequencies does most of the variation occur?* A white noise process spreads its variance evenly across all frequencies and produces a flat periodogram, while trends and persistent cycles create a peak at low frequencies. As in the lecture, frequencies are measured in radians, $0 < \lambda \leq \pi$, and the periodogram is $I(\lambda) = \text{spec}/(2\pi)$.

4.1 Periodograms

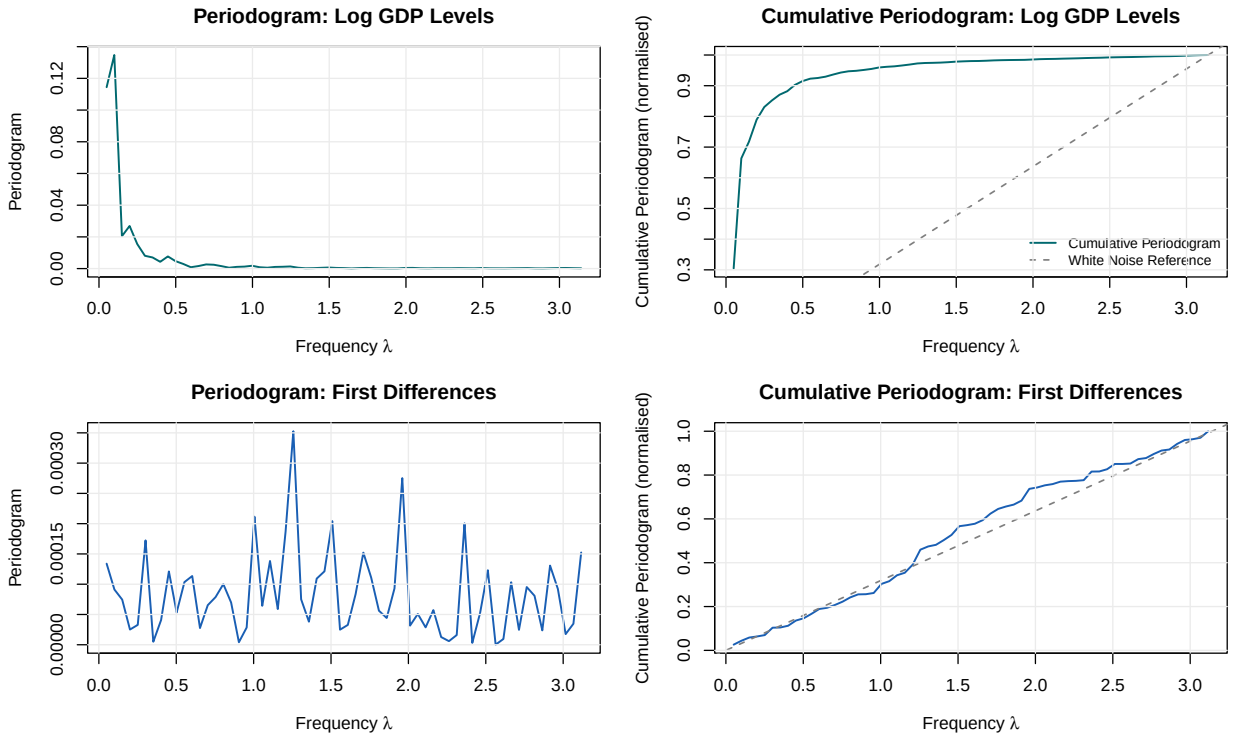


Figure 9: Periodograms (left) and cumulative periodograms (right) of log GDP levels (top) and first differences (bottom). Under white noise, the cumulative periodogram follows the diagonal dashed line.

For the log GDP levels (top row), the variance is concentrated almost entirely at the lowest frequencies: the three lowest frequencies alone account for 71.8% of the total. This is the spectral signature of the dominant trend. After first differencing (bottom row), the picture changes completely. The periodogram has no dominant peak, and the cumulative periodogram stays close to the diagonal white noise line. The formal test below checks whether the remaining deviations are larger than chance.

4.2 White Noise Test

Table 2: White noise test for differenced Croatian log GDP: sum of the periodogram at the j lowest frequencies times $4\pi/s^2$, compared with the 95% quantile of the χ^2_{2j} distribution.

j	Frequency (j)	Test Statistic	Chi-sq Quantile (95%)	p-value
10	0.5027	18.16	31.41	0.5767
20	1.0053	37.59	55.76	0.5791
30	1.5080	70.25	79.08	0.1717
40	2.0106	92.16	101.88	0.1664
50	2.5133	105.48	124.34	0.3345

The test checks whether too much variance is concentrated at low frequencies, which would mean that growth rates are persistent and partly predictable. Across all 5 thresholds the p-values lie between 0.166 and 0.579. **The white noise hypothesis is not rejected: quarterly GDP growth shows no significant serial structure.**

Two robustness checks point in the same direction. A Ljung-Box test with 8 lags gives a p-value of 0.602, and an AR(1) model for growth rates has a coefficient of only 0.059 (standard error 0.089). Excluding the extreme COVID-19 years 2020 and 2021 does not change the result (smallest p-value 0.283). In practical terms: knowing last quarter's growth rate does not help to predict next quarter's growth rate.

One limitation should be kept in mind. With 125 observations the test has little power against a slow change in the *average* growth rate, which concentrates at the very lowest frequencies. Such a change is visible in the data: mean growth before 2009 (0.88% per quarter) and from 2009 to 2019 (0.13%) differ significantly (Welch t-test, p-value 0.019). The white noise result therefore means that growth rates are not autocorrelated, not that their mean was constant over three decades.

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5 ARFIMA Long-Memory Model

5.1 Model Selection

Table 3: ARFIMA model comparison for quarterly growth rates of Croatian real GDP, estimated by profile likelihood. Parameters: d, the AR/MA coefficients and the innovation variance. The last column is the p-value of a likelihood ratio test of d = 0. The bold row marks the specification preferred by AIC.

Model	d	Log-Likelihood	Parameters	AIC	p (d = 0)
ARFIMA(0,d,0)	**-0.010**	**295.87**	**2**	**-587.74**	**0.894**
ARFIMA(1,d,0)	-0.126	296.57	3	-587.14	0.320
ARFIMA(0,d,1)	-0.122	296.78	3	-587.56	0.259
ARFIMA(1,d,1)	-0.111	296.80	4	-585.59	0.317

The ARFIMA model measures long-run memory through the parameter d. The model requires a stationary series ($-0.5 < d < 0.5$), so it is fitted to the quarterly growth rates rather than to the level of log GDP; the memory of the level is then $1 + d$.

The models are estimated by profile likelihood: for each candidate value of d the growth rates are fractionally differenced and an ARMA model is fitted by exact maximum likelihood, and d is chosen where this likelihood is highest. This approach also allows negative values of d. The standard routine `fracdiff` was not used for the comparison, because it only searches $0 \leq d < 0.5$ and its optimiser returned a lower likelihood for the models with AR or MA terms than for the simpler nested model, which is impossible at the true maximum.

Four specifications are compared by AIC (lower is better), and the preferred model is **ARFIMA(0,d,0)** with $\widehat{d}$ = -0.01. The simplest specification wins, so the data give no reason to add AR or MA terms. In none of the four models is d significantly different from zero (likelihood ratio p-values between 0.259 and 0.894). The semiparametric `fdSperio` estimator, which does not depend on a parametric model, agrees: $\widehat{d} = 0.031$ with a standard error of 0.038 (p-value 0.410). **The growth rates show no long memory.** For the level of log GDP this means d close to 1: Croatian GDP behaves like a random walk with drift.

5.2 Residual Diagnostics

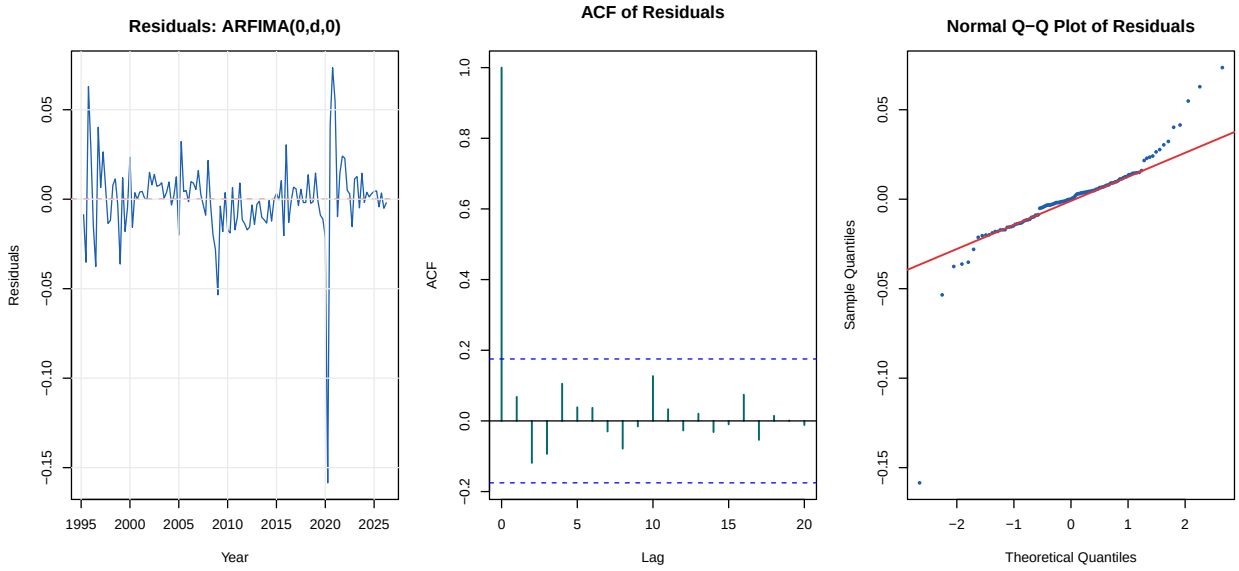


Figure 10: Diagnostic plots for the preferred ARFIMA model: residuals over time (left), autocorrelation function with 95% bands (centre) and normal Q-Q plot (right).

In the ACF, 0 of 20 autocorrelations fall outside the 95% bands (about one would be expected by chance), so no relevant serial dependence remains. The residuals are, however, clearly not normal (Shapiro-Wilk p-value < 0.001). The reason is visible in the left and right panels: the COVID-19 quarters produce extreme outliers, with the largest residual in 2020 Q2. A model for normal times cannot capture a shock of this size, which is a limitation of any linear time series model applied to this period.

5.3 Parametric vs. Nonparametric Spectrum

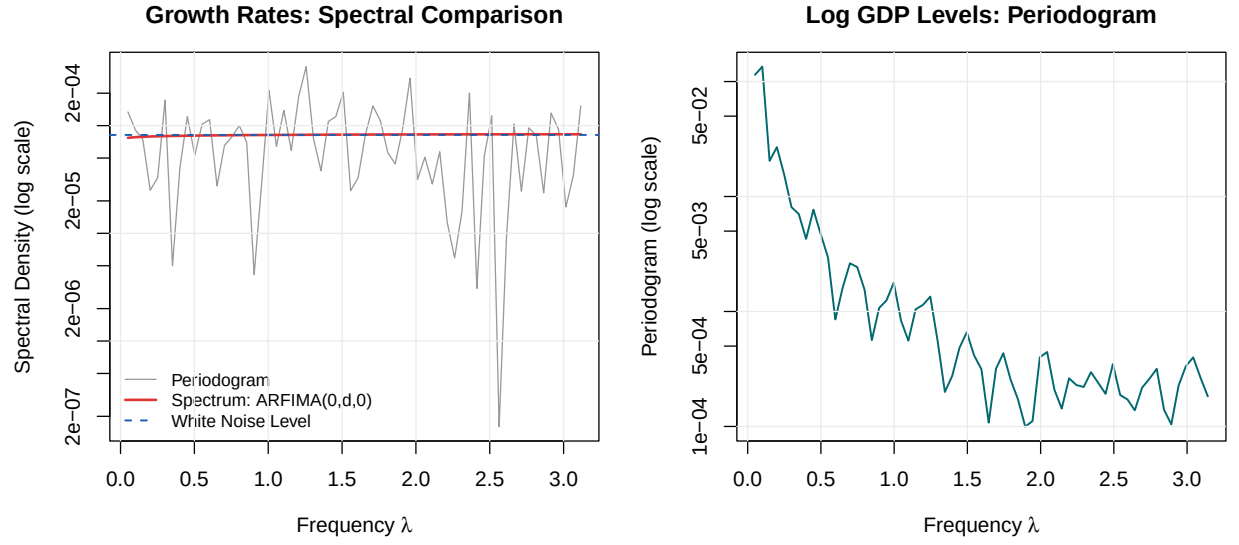


Figure 11: Left: periodogram of GDP growth rates (grey) with the spectral density implied by the preferred ARFIMA model (red) and the white noise level (blue, dashed). Right: periodogram of log GDP levels on a log scale.

For the growth rates (left), the spectral density implied by the ARFIMA model is flat and coincides with the white noise level: the periodogram scatters around this line without any systematic shape. For the levels (right), the periodogram falls by several orders of magnitude from the lowest to the highest frequencies, the typical picture of an integrated series. Together, both panels describe Croatian GDP as a trending level whose quarterly changes are essentially unpredictable.

6 Conclusion

This analysis of Croatian real GDP from 1995 Q1 to 2026 Q2 leads to a clear picture.

The long-run path is shaped by two crises. The 2008 Global Financial Crisis lowered trend growth from 0.80% to 0.01% per quarter, and the pre-crisis peak was only regained after 10.5 years. The COVID-19 shock was far deeper in the short run, but GDP returned to its pre-pandemic level within 3 quarters.

Quarterly growth is not predictable from the past quarters. The spectral white noise test from the lecture, a Ljung-Box test and an AR(1) model all find no significant autocorrelation in the growth rates, and the ARFIMA memory parameter of the growth rates is not significantly different from zero.

Shocks are permanent, and the drift can shift. Taken together, Croatian log GDP behaves like a random walk with drift: each shock shifts the level of output for good instead of fading away. The drift itself was not constant, however. Average growth fell from 0.88% per quarter before 2009 to 0.13% between 2009 and 2019 and recovered to 0.91% afterwards. This is why the 2008 crisis left such a lasting mark, and why the structure of Croatian GDP lies in its trend and structural breaks rather than in short-term growth dynamics.

This analysis was created by Filip Popovic as part of a data analyst portfolio. For questions or inquiries, ***contact me at fpopovic01@gmail.com****.*